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B.E - (Computer) (Semester-VII)(Revised Course 2019-2020)**CE 723 Data Analytics (PE)****[Time: 3:00 Hours]****[Max. Marks:100]**

- Instructions:**
- 1) Assume suitable data if necessary
 - 2) Answer any two questions from Part A and Part B. Also attempt one question from Part C.
 - 3) Draw neat diagrams if required.
 - 4) Write question numbers legibly while answering.
 - 5) Write description for the questions based on the marks allotted.
 - 6) Use of statistical tables permitted if required

Part A

- Q1**
- A. What are the benefits of pruning in decision tree induction? Explain different approaches to tree pruning with illustration **8**
 - B. State the mathematical formulation of the SVM problem. Give an outline of the method of solving the problem. **6**
 - C. Discuss Major steps involved in the data preprocessing to make sure that, data is suitable for Analytics. **6**
- Q2**
- A. Briefly, describe the importance parametric tests used in context of testing of hypotheses. How such tests are different from non-parametric tests? Explain. **8**
 - B. Eight coins were tossed 256 times and the following results were obtained **6**

No of Heads	0	1	2	3	4	5	6	7	8
Frequency	2	6	30	52	67	56	32	10	1

Are the coins are biased? Use Chi Squared Test.

- C. Explain the types of Correlation with example **6**
- Q3**
- A. Explain the meaning of analysis of variance. Describe briefly the technique of analysis of variance for one-way classification. **8**

- B. Apply linear regression using method of least squares to the following data and predict crop yield for rainfall of 5cm. **6**

Rainfall in cms	10.5	8.8	13.4	12.5	18.8	10.3	7.0	16
Crop yield (quintal per acre)	30.3	46.2	58.8	59.0	82.4	49.2	31.9	78.8

- C. Illustrate k-Medoids clustering algorithm, and how does it differ from k-Means **6**

Part B

- Q4** A. Explain the different activities performed in Discovery phase of Data Analytic Life Cycle. **8**

- B. Describe responsibilities, skills and deliverables of the following roles in successful data analytic projects. **6**

- Data Analyst
- Data Scientist
- Data Engineer

- C. Describe the components of the DBMS. **6**

- Q5** A. Explain briefly different types of NOSQL databases with example systems. **8**

- B. List and explain different levels of views supported in database systems **6**

- C. Give the comparative features of SQL and NOSQL based database systems. **6**

- Q6** A) Differentiate between scalars, vector, list, matrix and data frames in R language with suitable example. **8**

- B) Write R programming code to **6**

- I. Create an Empty Data Frame with Various Column Types in R
- II. Create a Data Frame from four Vectors
- III. Print the statistical summary of the data frame created in step II

- C) Illustrate the functions which helps in importing data from other applications in R language with an example. **6**

Part C

- Q7** A). Discuss the significance of visualization and R's support for it **8**
 B). Describe various functionalities supported for data exploration in R. **6**
 C). Explain the difference between probability and non-probability sampling? **6**
- Q8** A). Consider the following set of training examples given below **10**

Instance	Classification	A1	A2
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

- (i) What is the entropy of this collection of training examples with respect to the target function "Classification"?
- (ii) What is the information gain of A2 relative to these training examples?
- B). Briefly explain commonly used tools for the Model Building Phase and Also list out Key outputs from a successful analytics project **10**